

Question Paper

Exam Date & Time: 03-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Distributed Systems [CSE_3251]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Define a distributed system and identify its main characteristics. (2)
A) Course Outcome: "CO1" / Cognitive Level: "Application"
B) Show the different middleware organizations? Give examples. (3)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Illustrate the layered architectural design of distributed systems with examples. (5)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Organize the essential foundations of distributed communication. (2)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Show the working of Remote Procedure Call (RPC). (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Compare and contrast message-oriented communication and multicast communication. (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Identify the components of a naming system? Explain briefly with suitable examples. (2)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Give the advantages and disadvantages of flat naming. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Infer the use of hierarchical naming used in distributed systems? (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Demonstrate the concept of synchronization in distributed systems. (2)
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) Illustrate the vector clocks and how they solve causal ordering. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Describe the Bully election algorithm with an example. (5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

5) Show replica management with suitable example. (2)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Identify the difference between strong and eventual consistency. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

C) Discuss consistency protocols used in large-scale AI data centres. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3251:Distributed Systems

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3251:Distributed Systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	19(38%)
4	Analysis	5	21(42%)
5	Create	2	10(20%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)